



BUILT FOR HANDS THAT ARE BUSY

# Gloves on. Hands full. Just speak.

Speak a goods receipt. Review and confirm it. It posts into **SAP** — a mock S/4HANA behind SAP's released OData API.

Goods receipt · Stock · Reversal · SAP OData

AssemblyAI Voice Agent Hackathon · September 2026



RECEIVING DOCK • MORNING SHIFT

# Lena has both hands under a box.

The pallet landed four minutes ago and it is on no one's books yet. Until that delivery is a document in SAP, the company does not know it owns the goods — purchasing chases it, production plans without it.

**Booking it costs Lena a walk: gloves off, across the floor, to a shared terminal. Every delivery, every shift.**

WHAT THIS TOUCHES — AND WHAT IT DOES NOT

# A shift is a bundle of tasks.

|                             |           |                                 |           |
|-----------------------------|-----------|---------------------------------|-----------|
| Unload the pallet           | UNTOUCHED | Type the material number        | GONE      |
| Check the packing slip      | UNTOUCHED | Look up which bin it goes to    | AUTOMATIC |
| Set damaged cartons aside   | UNTOUCHED | Post the goods receipt          | BY VOICE  |
| Take the gloves off         | GONE      | Correct a wrong posting         | BY VOICE  |
| Walk to the shared terminal | GONE      | Hear the line before it is true | NEW       |

**GlovesOn cannot unload the pallet. It removes the three tasks that exist only because the record lives on a screen — and adds one worth doing.**

SAY IT. IT POSTS.

# Both hands on the box. The receipt **posts into SAP.**

YOU "Twenty pieces of material 4711 arrived."  
GLOVESON "Twenty pieces of Hex Bolt M8x40 Zinc Plated into bin A-03-02 – confirm?"  
YOU "Actually twelve."  
GLOVESON "Twelve pieces ... into bin A-03-02 – confirm?"  
YOU "Yes."

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→ **Material document 4937706921 · movement 501 · stock 271 → 283**

**A mock S/4HANA that speaks SAP's released OData contract. One setting from a live tenant. Hands never leave the box.**

REAL SCREEN · SESSION OF 15 SEPTEMBER 2026

# A result you can review.

GOODS RECEIPT · MATERIAL DOCUMENT · MOCK S/4HANA

Document **4937706921** ✓ assigned 13:51:56

RECORDED · 13:51:56

stock 271 → 283 EA

**12 EA of Hex Bolt M8x40 Zinc Plated are in bin A-03-02.**

Wrong? Say "reverse it". A reversal takes them back out and both documents stay.

Material ✓  
**4711**

Description ✓ material master  
**Hex Bolt M8x40 Zinc Plated**

Quantity ✓ confirmed  
**12 EA**

Storage bin ✓ the material's own  
**A-03-02**

Plant / location ✓ default  
**1000 / 0001**

Stock ✓ from the same result  
**271 → 283 EA**

WHY THIS DOCUMENT EXISTS  
"Twelve pieces of Hex Bolt M8x40 Zinc Plated into aisle A, level three, position two." — as the agent reported it, 13:51:56.

✓ Receipt saved

Material document **4937706921**, movement **501**, stock **271 → 283** — recorded through SAP's released OData path into the demo's mock S/4HANA, after the worker corrected twenty to twelve during the read-back.

The read-back names the **description**, not just the number — which is what lets a worker catch they are holding nuts, not bolts.

The card carries **the sentence that caused the document**. An earlier run, document 4922857164 with its refused duplicate and its reversal, is kept as JSON in receipts/.

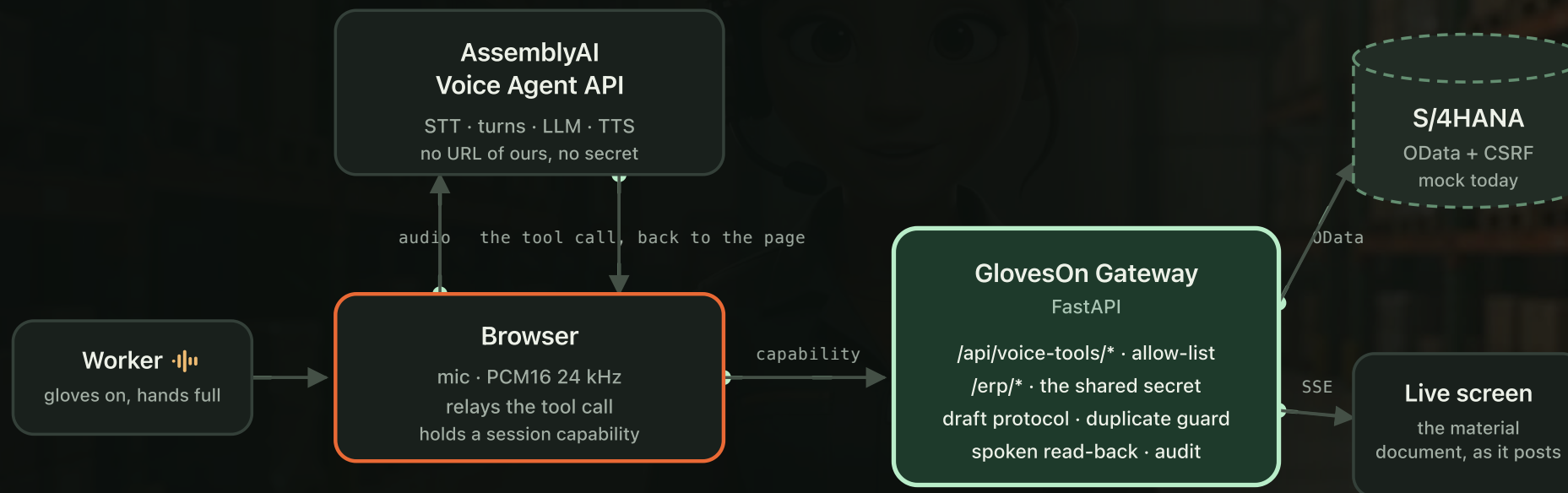
ASKING IS EASY. FINISHING IS THE HARD PART.

# Five guardrails, each one deliberate.

- 1 The read-back names four things.**  
Quantity, unit, material description, bin. The description is load-bearing.
- 2 Writes require a confirmed draft.**  
The gateway validates the exact draft. Spoken consent is reported by the agent; results appear only after the tool returns.
- 3 A duplicate is refused.**  
The same receipt within two minutes is caught — by a guard in the gateway, because real S/4HANA would accept it twice.
- 4 Nothing is deleted — it is reversed.**  
A wrong receipt gets a reversal document, 102 against a 101. Both stay, because that is what a ledger is.
- 5 Every document answers why it exists.**  
Click a posted row: the sentence the worker confirmed, and the second it happened.

THE AGENT NEVER SEES SAP

# One boundary carries every guardrail.



the agent never sees SAP, and never sees the gateway either — the write is the gateway's alone

It speaks in warehouse terms; the gateway owns auth, SAP field translation and every write guardrail. The mock sits opposite the gateway, reached like any other system — so **SAP\_BASE\_URL** pointed at a real tenant is the whole migration. No second code path.

THE WHOLE VOICE LOOP IS THE VOICE AGENT API

# One connection: speech, reasoning, voice, tools.

Streaming **speech-to-text**

LLM routing

Voice output

Turn-taking & VAD

**JSON-Schema** client functions

Browser-to-gateway tool bridge

transcription\_prompt

39 keyterms → SAP material numbers

voice\_focus: far-field

Barge-in

session.resume

Session token minted server-side

Turn detection is pinned (**800 / 2000 ms**) for non-native speakers — adaptive pacing is deliberately off, so we do not claim per-speaker adaptation.

EVERYONE ELSE VOICE-GUIDES PLANNED WORK

# The unplanned pallet has no voice path.

## THE INCUMBENTS

### Honeywell · Lydia · Ivanti

The WMS issues the task; the worker answers fixed prompts. Decades of it. But a worker cannot initiate a document by describing it.

## SAP ITSELF

### Joule

Posts and reverses goods movements — by **chat**, today. Real-time voice announced for its top cloud tiers, GA planned H2 2026.

**GlovesOn is the unplanned moment — by voice, today.**

AN ESTABLISHED BUDGET LINE

# Zero per-seat hardware against a \$5,000 norm.

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## \$4.8–6.5B

voice-directed warehousing today · ~14–17%/yr (analyst estimates, cited as a range)

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## ~\$5,000

incumbent cost per user, hardware + software

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## \$0.00

GlovesOn per-seat hardware — a browser and a headset. Voice: \$0.075 per connected minute.

No ROI figure is quoted — session length per receipt is unmeasured (docs/nfr.md §5).

NAMED, NOT HIDDEN

# What this does not do yet.

- **The ERP is a mock** — the OData contract and the CSRF handshake are real; a live tenant was tried and the Business Accelerator Hub sandbox answered **401** on all four required services (10 September). Nothing here has run against real S/4HANA.
- **No employee sign-in or SAP principal propagation** — session capabilities protect tool calls, but do not establish a worker's identity.
- **Noise robustness is unmeasured** — voice\_focus: far-field is set, but a config change is a claim, not a measurement.
- **Session events are isolated; warehouse records are shared** — this is a demonstration, not a tenant-isolated production system.
- **The supported product language is English**; other languages are outside the demo scope.

Naming the gap is the credibility. Every one is written up in the repo, and the judge guide ends with them.

## WHAT A JUDGE CAN CHECK

# Posts the goods receipt **straight into SAP** — here a mock S/4HANA that speaks SAP's released OData contract, one setting away from a live tenant.

## WHAT IT DOES

- `X-CSRF-Token`: Fetch handshake, `A_MaterialDocumentHeader`, movement types 101 / 102
- Read-back names quantity, unit, material *description* and bin; nothing posts without a spoken yes
- Wrong receipt is reversed, never deleted — both documents stay
- The same receipt twice is refused in the gateway

## WHAT YOU CAN CHECK

- **gloveson.space** — no login, talk to it
- `github.com/kosesena/GlovesOn` — **281 commits** since 5 Sept, **97 tests**, CI green, MIT
- `receipts/` — **13 steps** of one real run: document **4922857164** posted, the same receipt refused, then reversed
- `/api/provenance/4922857164` — the sentence that caused the document
- `docs/adr/` — six decisions, each with the option rejected and its cost

## NOT VERIFIED

- Real S/4HANA: the sandbox answered **401** on all four required services
- The posting is not made as the worker — no principal propagation yet
- Accents and forklift noise: not benchmarked
- Email, calls and notes are demo records; nothing reaches a real person

GLOVESON

# It doesn't just stop and ask. It finishes — and leaves a document that can tell you **why it exists.**

Live demo · [gloveson.space](https://gloveson.space)

[github.com/kosesena/GlovesOn](https://github.com/kosesena/GlovesOn)

Judge guide — every claim → a file & a command

Gloves on. Hands full. **Just speak.**

Built on the AssemblyAI Voice Agent API

Demo: mock ERP · simulated calls · demo email outbox